



Edouard Pignède

Curriculum Vitae

Current position

2025–Present **Post-doctoral scholar**, *French National Research Institute for Sustainable Development (IRD), Université Paris-Saclay, UMI SOURCE*, Paris,
Member of the BETSAKA project, initiative to assess the environmental and socio-economic impacts of protected areas in Madagascar from 2000 to 2024

Academic appointments

2025 **Research Fellow**, *NovAfrica (NOVA SBE)*, Lisbon,
Member of the FCTPex project *Artisanal Gold Mining and Sustainable Development*

Education

- 2021–2025 **PhD, Development and environmental economics**, *AgroParistech (Paris-Saclay University) & Climate Economic Chair*, Paris,
Climate change adaptation in Sub-Saharan Africa: Inequalities and vulnerabilities
Chapter 1: impact of drought on inequalities using quantile treatment effect and distribution regression, as well as the initial endowment in capital.
Chapter 2: empirical analysis of the link between artisanal mining and urbanization.
Chapter 3: quantitative spatial modeling of climate change-induced migration with liquidity constraints.
Chapter 4: quantitative spatial modeling of climate change impact on mortality.
- 2019–2020 **Master's degree in environmental economics**, *AgroParistech (Paris-Saclay University)*, Paris,
Average: 15.95
Courses: environmental econometrics, economy of natural resources, international trade and environment, public policies and environment, climate change economy
- 2016–2020 **Master's degree in industrial engineering**, *CentraleSupélec (Paris-Saclay University)*, Paris,
Average: 16.35
Specialization in industrial engineering and management of international project.
Courses: advanced probability, advanced statistics, economy, sustainable development, modelling and optimization.
Six-month project on applied mathematics: development of modelling algorithm of Hawkes process computed with Python.
- 2014–2016 **Classes préparatoires**, *Henri IV*, Paris
A two-year intensive program in mathematics and physics to prepare for nationwide competitive entrance in top French graduate schools of engineering.

Publications

Journal Articles

- 2025 **Edouard Pignède**. Who carries the burden of climate change? Heterogeneous impact of droughts in sub-Saharan Africa. *American Journal of Agricultural Economics*, 2025. <https://doi.org/10.1111/ajae.12507>.
- 2021 **Edouard Pignède**, Philippe Roudier, Arona Diedhiou, Vami Hermann N'Guessan Bi, Arsène T. Koba, Daouda Konaté, and Crépin Bi Péné. Sugarcane yield forecast in Ivory Coast (West Africa) based on weather and vegetation index data. *Atmosphere*, volume 12, page 1459, 2021. <https://doi.org/10.3390/atmos12111459>.

Non-refereed journal articles

- 2026 Victoire Girard and **Édouard Pignède**. Artisanal Mining and Urbanization in Africa. *Nova SBE Working Paper Series*, 2026.
- 2021 **Edouard Pignède** and Guillaume Daudin. French Trade in the 18th century: Inflation and Revolution. *Dialogue*, 2021. English version: [link](#). French version: [link](#).

On-going works

- 2026 **Edouard Pignède**, Florent Bedecarrats and Thomas Thivillon, Protected areas and fires in Madagascar
- 2025 **Edouard Pignède**, Climate change, adaptation, and mortality
- 2024 **Edouard Pignède**, and Julien Wolfersberger, Climate immobility in sub-Saharan Africa, Available upon request

Research Experience

- Nov,2020 – **Research engineer**, *French National Research Institute for Sustainable Development (IRD)*,
Nov,2021 *DIAL*, Paris
- Mapping of Sub-Saharan vulnerability to climate change through three components: exposition, sensibility and adaptation capacities. Building of multiple indicators of vulnerability using household survey (DHS, agricultural survey), satellite data (CHIRPS, ISRIC) and country data (infrastructures, mines localisation) computed with RStudio and QGIS.
- Application of an original statistical method of spatial matching to assess the effect of roads on inter-sectoral mobility and population migration in Mali, using national census data and RStudio.
- Building a global database of coast escarpment, nightlight and wave height to assess the role of piracy on coast development using RStudio and QGIS.
- Computing of price indexes in France in the 18th century using the repeated sales method, RStudio and data from the TOFLIT project (published in Dialogue n°62, English version: [link](#). French version: [link](#)).
- May,2020 – **Research engineer**, *French Development Agency (AFD)*, Paris, End-of-study internship
Nov,2020
- Development of statistical models of prediction of sugarcane yields in Ivory Coast based on machine learning algorithm (random forest) and using meteorological data (precipitations, temperatures) and satellite data (NDVI, EVI). Computation with RStudio and QGIS.
- Publication of a scientific paper in a peer-reviewed journal as well as a research paper in French.

Conference and talks

- 2026 CLAND Workshop - Land Use Policies for Climate Change, Paris (France)
ADRES - Job Market Conference, Paris (France)
Seminar of UMI SOURCE, University of Paris-Saclay

- 2025 FAERE - Conference of the French Association of Environmental Economists, Nantes (France).
EEA Congress, Bordeaux (France).
ICDE - International Conference in Development Economics, Nanterre (France).
CEEM Seminar, Agro Institute of Montpellier - INRAE.
CSAE Conference in Development Economics, Oxford (England).
Lisbon Micro Group, ISEG, Lisbon (Portugal).
DIAL Seminar, Paris-Dauphine University.
- 2024 EAERE - Conference of the European Association of Environmental Economics, Leuven (Belgium).
NOVAFRICA Conference on Economic Development, Lisbon (Portugal).
Paris-Saclay Conference on Trade and the Environment, University Paris-Saclay
CSAE Conference in Development Economics, Oxford (England).
Workshop in Environmental Economics, Toulouse School of Economics (France).
- 2023 PhD Research Group of NOVAFRICA, Lisbon (Portugal).
Lisbon Micro Group, ISEG, Lisbon (Portugal).
FAERE - Conference of the French Association of Environmental Economists, Montpellier (France).
EAAE - European Association of Agricultural Economists Congress, Rennes (France).
ICDE - International Conference in Development Economics, Paris (France).
AFSE - Congress of the French Economic Association, Paris (France).
Paris-Saclay Applied Economics Internal Seminar, Paris (France).
EAERE Winter School on Natural Resources and Development, Ascona (Switzerland).
- 2022 AgEconMeet - Early career seminar for agricultural economists in Europe, Göttingen (Germany).

Skills

- Computer skills Advanced knowledge of R, Python and QGIS
Environment: Quarto, Positron, SSP Cloud, GitHub, Zotero
Basic knowledge of STATA
- Languages French: Native
English: Fluent
Spanish: Intermediate

Teaching experience

- 2026 **Training in research replicability**, *PhD Program - CERED (University of Antananarivo)*, Madagascar
Taught in French. 14-hour course at the PhD level.
- 2023 **Training in research replicability**, *PhD Program - PSAE (Paris-Saclay University)*, Paris
Taught in French. 2-hour course at the PhD level.
- 2023 **Introduction to Environmental Economics**, *Master BIOCEB - AgroParistech (Paris-Saclay University)*, Paris
Taught in English. 9-hour course at the Master level.
- 2022 **Introduction to Climate Change in Economics**, *Master ADEPP - ENSAE Dakar*, Dakar
Taught in French. 2-hour course at the Master level.
- 2022 **Microeconomy**, *CPES - PSL University*, Paris
Taught in French. 20-hour course at the Bachelor level (2nd year).
- 2022 **On the use of GIS data in economy**, *AgroParistech (Paris-Saclay University)*, Paris
One session introduction lecture course to M2 students.
- 2018 **Teaching assistant in Information System**, *CentraleSupélec (Paris-Saclay University)*, Paris
Course for 1st year student.
- 2018 **Personal mathematics teacher**, Paris
Teaching mathematics to 10 high-school students.

Referee reports for scientific journals

- 2026 IPCC Special Report on Climate Change and Cities
2025 American Journal of Agricultural Economics
Agricultural Economics
Journal of African Economies
2024 American Journal of Agricultural Economics

References

Dr. Philippe Delacote

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